MILITARY STANDARD

CARLOADING

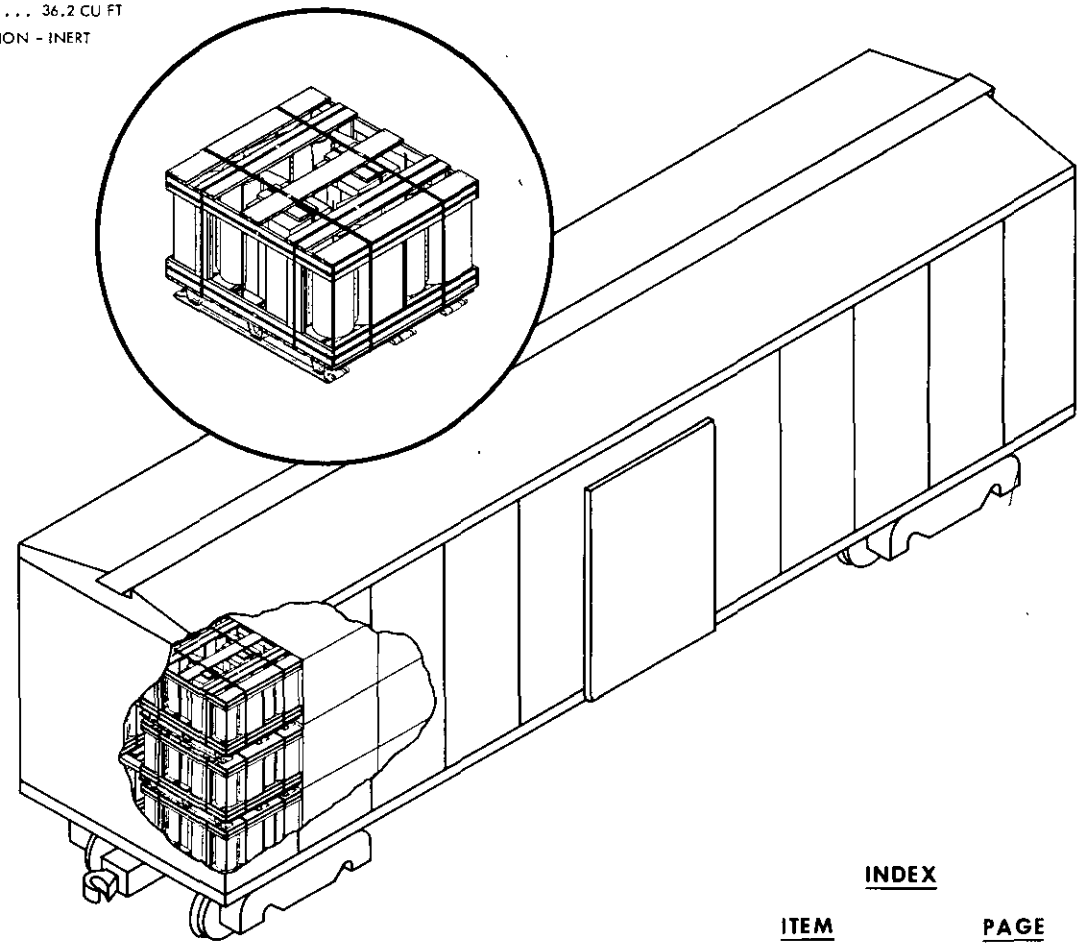
BOMB GUIDANCE KIT
KMU-388/B (LESS SEEKER ASSEMBLY)
FLEET ISSUE UNIT LOAD

MIL-STD-1325-162
(NAVY)

27 MARCH 1973

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/235
WEIGHT . . . . . 605 LBS
DIMENSIONS . . 48 L x 41 3/4 W x 31 1/4 H
CUBE . . . . . 36.2 CU FT
DOT HAZARD CLASSIFICATION - INERT



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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

AUTHORIZED AND RELEASED FOR GENERAL USE.

M. S. Gray NWHL 3/26/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 3/26/73
SIGNATURE (AIR) SYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES
NOT APPLICABLE
DATE

ORIGINATOR Charles McBride 3/26/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF BOMB GUIDANCE KITS, KMU-388/B (LESS SEEKER ASSEMBLY) PALLETIZED IN ACCORDANCE WITH WR-54/235.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY)
9. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC MM-L-751
   NAILS - FED. SPEC FF-N-105
   STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
   SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS.

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 78 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSS MEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

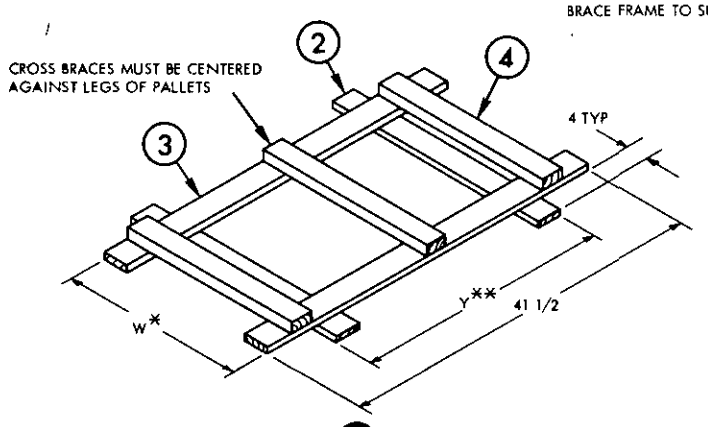
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
5	LATERAL FLOOR MEMBER	2 x 6 x 107	2	CAR FLOOR	2 PER FOOT	16d
4	CROSS BRACE	2 x 4 x W*	78	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 41 1/2	52	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	52	SEE 3	-	-
1	SLEEPER	2 x 4 x 40	26	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

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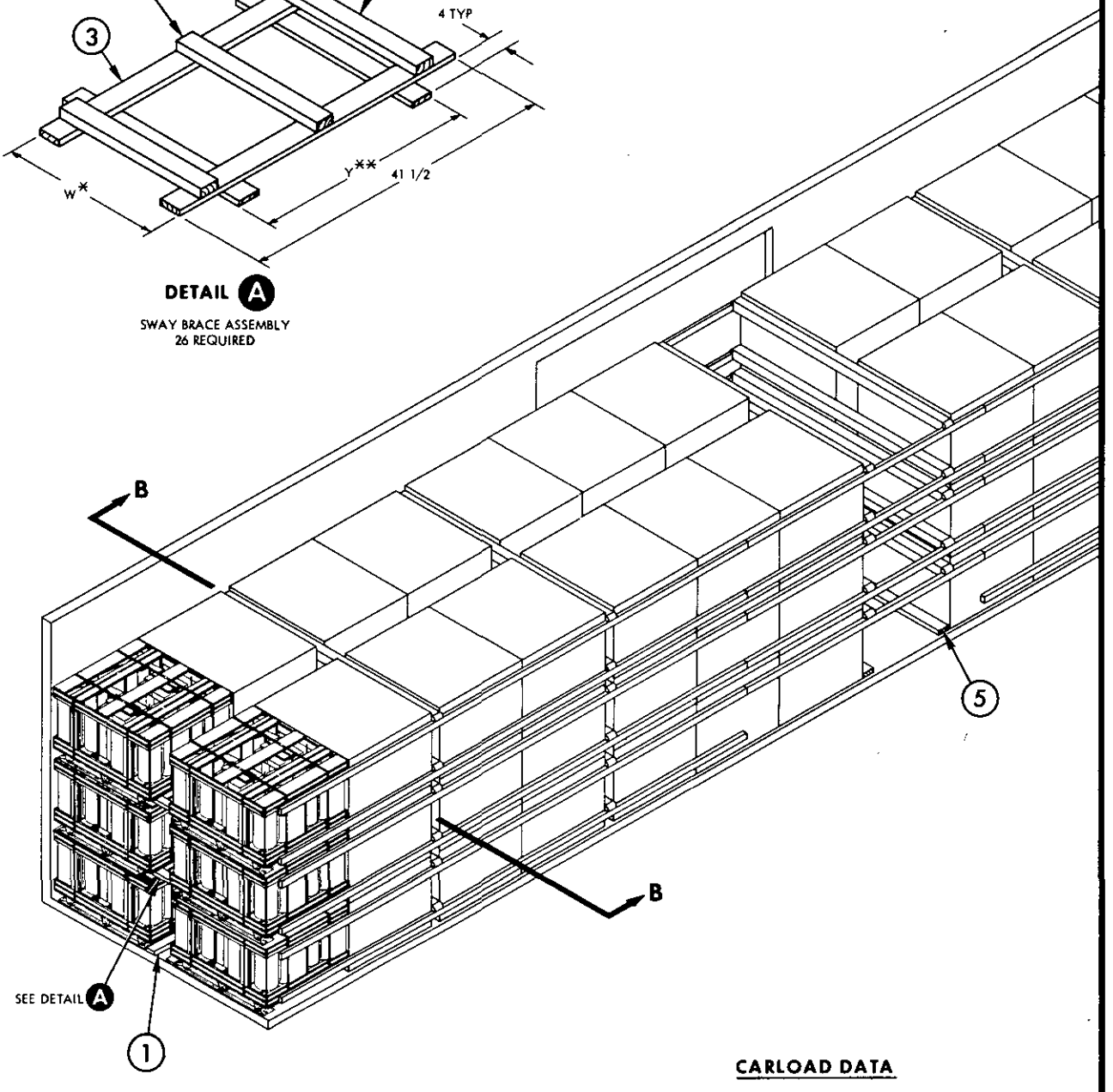
*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET. INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



DETAIL A

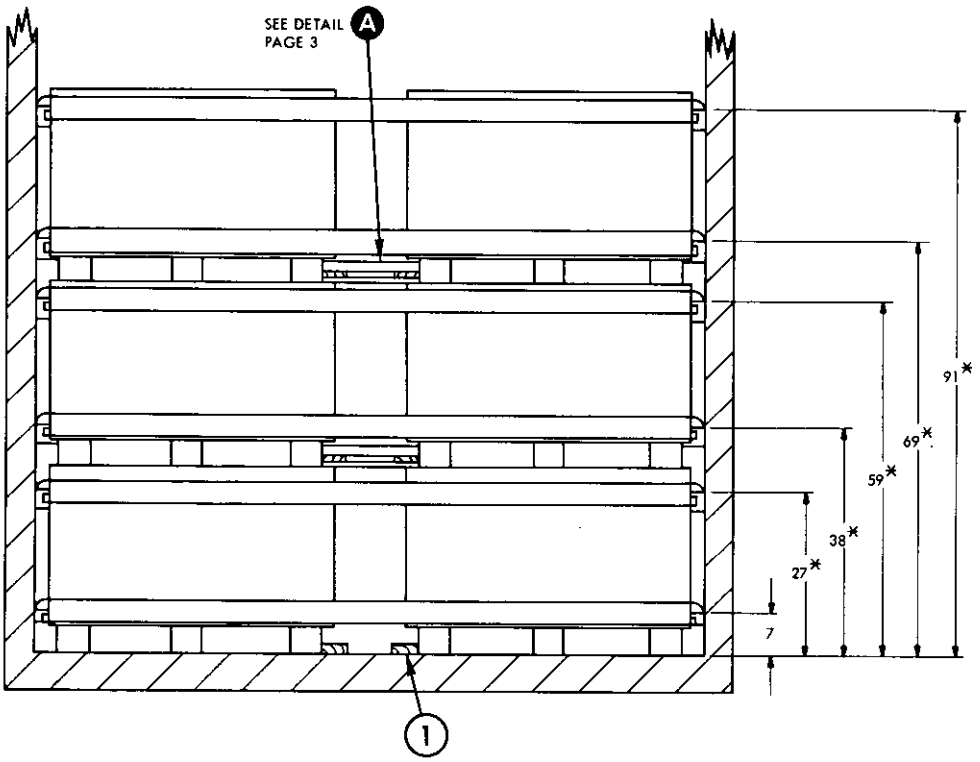
SWAY BRACE ASSEMBLY
26 REQUIRED



CARLOAD DATA

NUMBER OF UNIT LOADS	78
NUMBER OF CROSS MEMBERS REQ'D	34
LOAD WEIGHT	47,190 LBS
DUNNAGE WEIGHT	607 LBS
CARLOAD WEIGHT	47,797 LBS

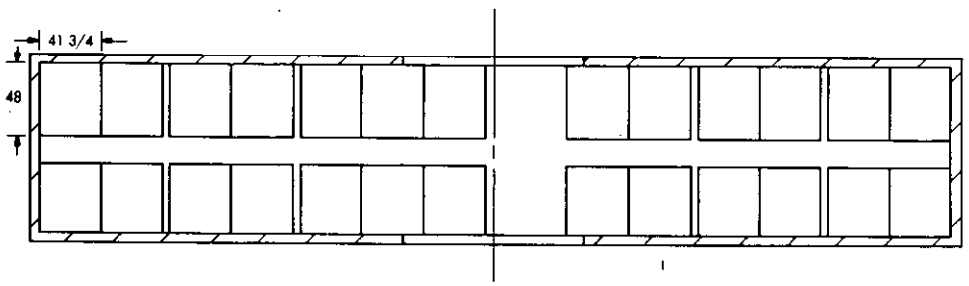
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SECTION B-B

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSS MEMBERS

* DOORWAY MEMBER LOCATIONS



LOADING PLAN

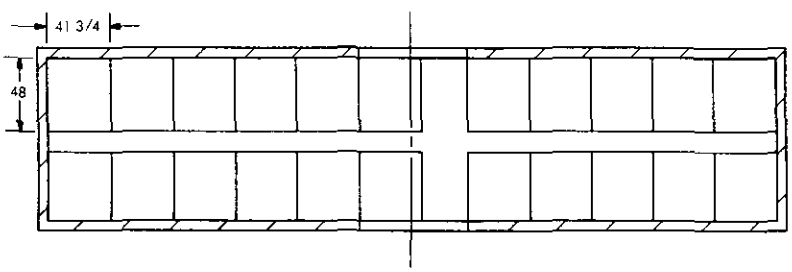
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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 66 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. *WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS,* THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH MIL-STD-1325 (NAVY).



LOADING PLAN

* CLINCHED

**W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

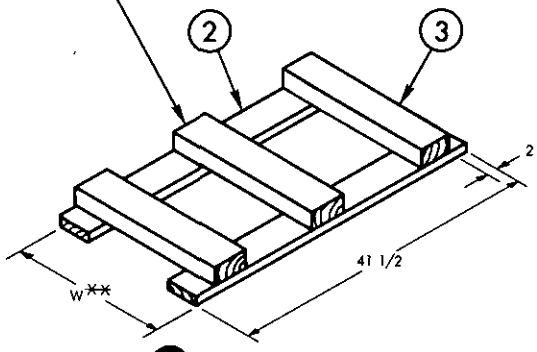
*** 2 x 6 STRUTS, DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
11	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	10	2 PER JOINT	10d
10	VERTICAL DOORWAY MEMBER	2 x 3 x 108	4	DOOR POST	2 PER FOOT	20d
9	HOLD DOWN CLEAT	2 x 6 x 48	2	8	2 PER FOOT	10d
8	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
7	STRUT	4 x 4 x WEDGE FIT***	24	5	2 EACH END	16d
6	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	12	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 x 6 x 96	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	12	5	3 PER JOINT	10d
3	CROSS BRACE	2 x 4 x W**	66	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 x 4 x 41 1/2	44	3	3 PER JOINT	10d*
1	SLEEPER	2 x 4 x 40	22	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

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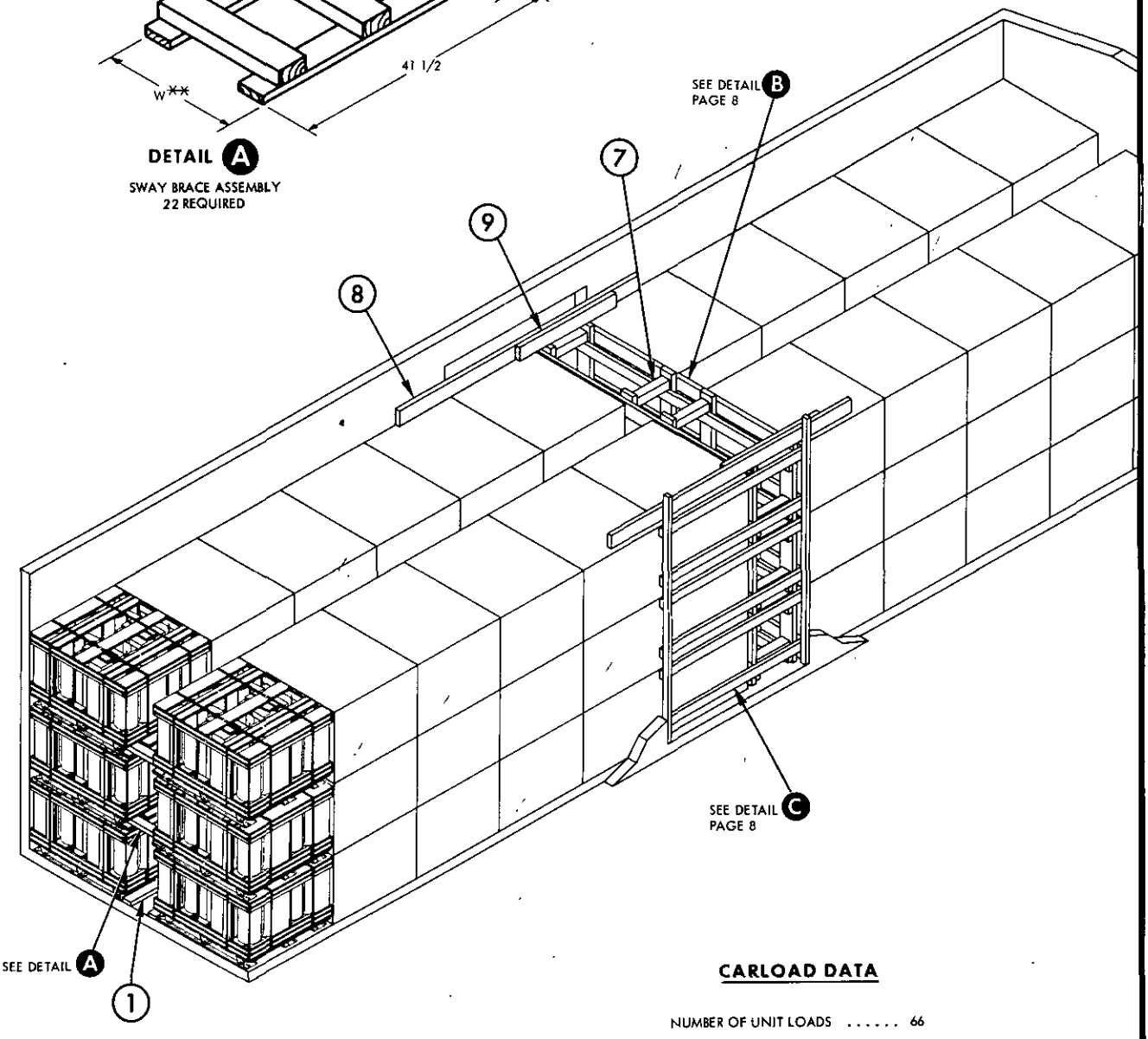
CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS



**W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

DETAIL A

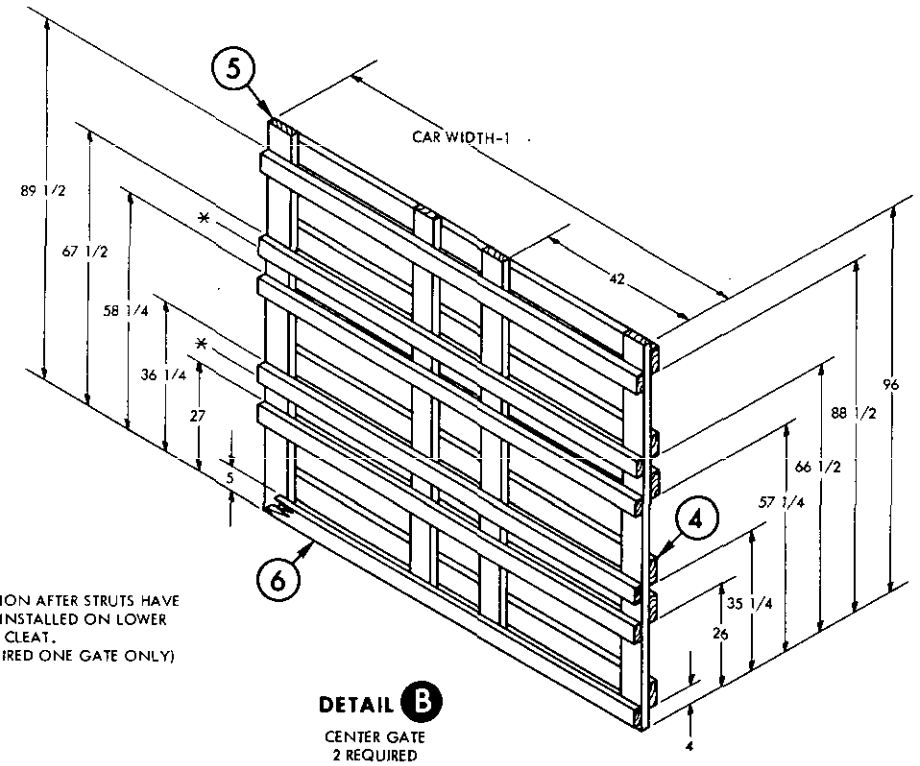
SWAY BRACE ASSEMBLY
22 REQUIRED



CARLOAD DATA

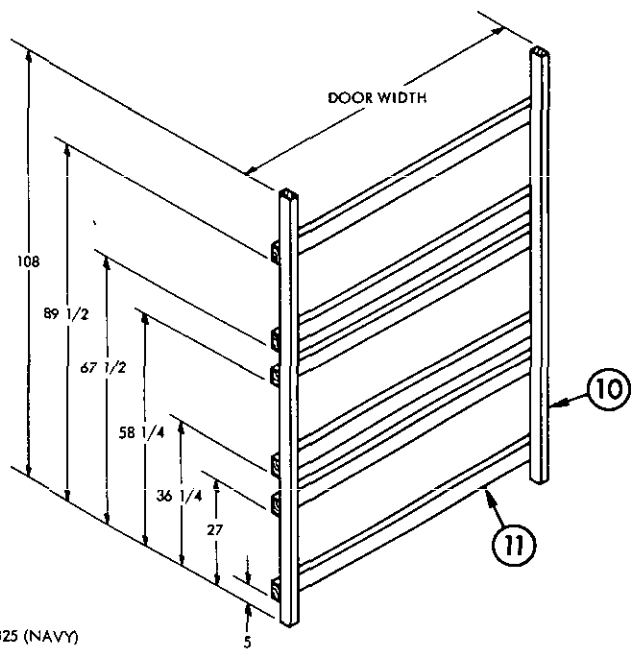
NUMBER OF UNIT LOADS	66
LOAD WEIGHT	39,930 LBS
DUNNAGE WEIGHT	1,467 LBS
CARLOAD WEIGHT	41,397 LBS

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*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL B
CENTER GATE
2 REQUIRED



WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

DETAIL C
DOORWAY PROTECTION
2 REQUIRED

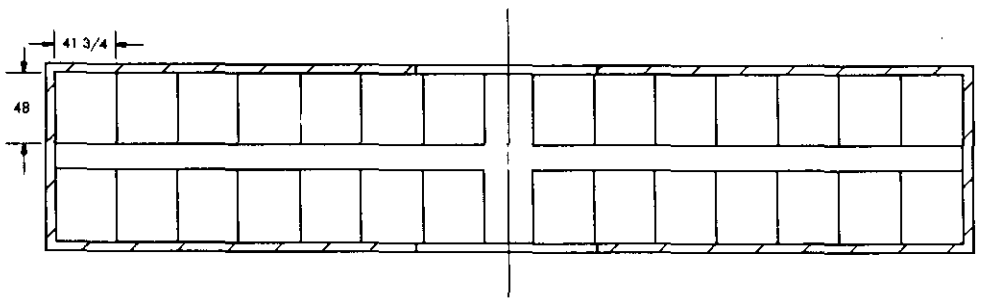
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 84 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACING TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 8 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 8 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND DOORWAY PROTECTION, PIECES 10 AND 11, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

* CLINCHED

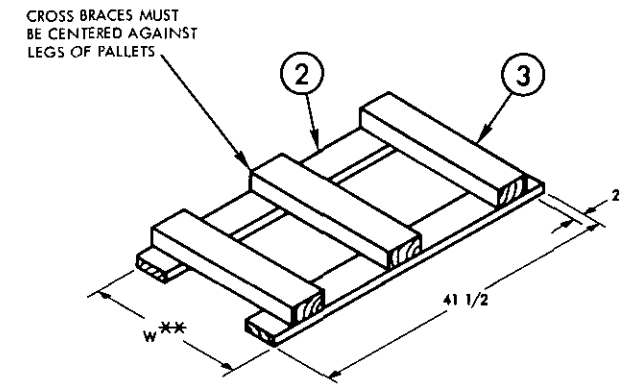
**W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

*** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
11	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	10	2 PER JOINT	10d
10	VERTICAL DOORWAY MEMBER	2 x 3 x 108	4	DOOR POST	2 PER FOOT	20d
9	HOLD DOWN CLEAT	2 x 6 x 48	2	8	2 PER FOOT	10d
8	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
7	STRUT	4 x 4 x WEDGE FIT***	24	5	2 EACH END	16d
6	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	12	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 x 6 x 96	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	12	5	3 PER JOINT	10d
3	CROSS BRACE	2 x 4 x W**	84	SEE 2	-	-
2	SWAY BRACE LONGITUDINAL	1 x 4 x 41 1/2	56	3	3 PER JOINT	10d*
1	SLEEPER	2 x 4 x 40	28	CAR FLOOR	2 PER FOOT	16d

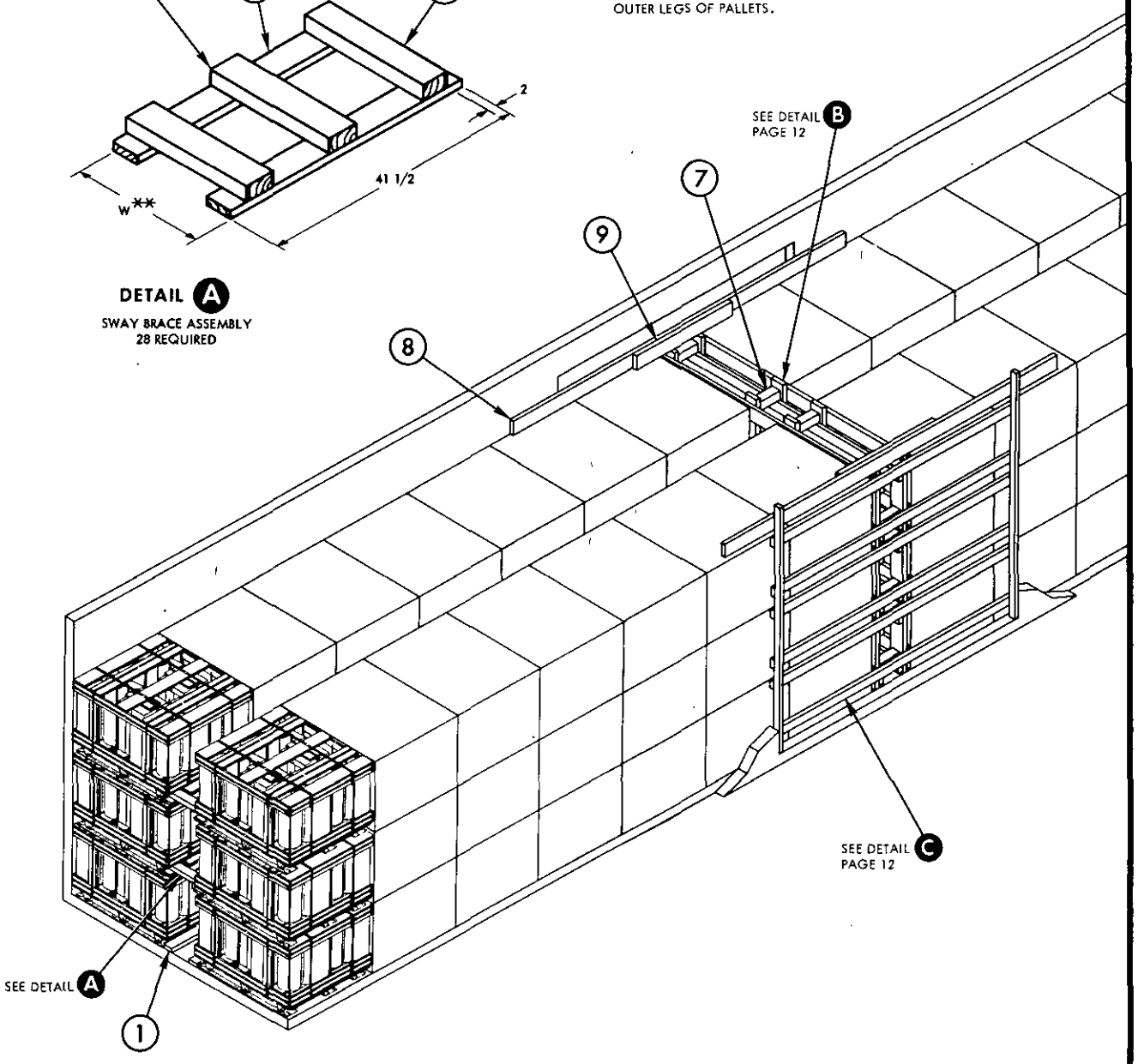
LIST OF MATERIALS AND NAILING DATA

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**W = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLETS.

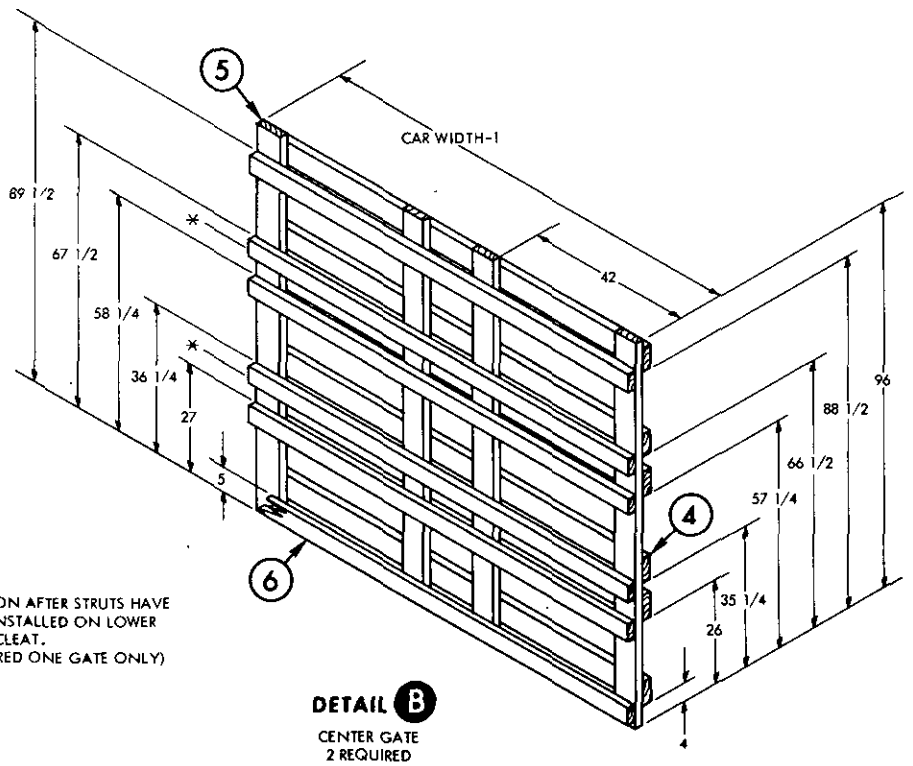
DETAIL A
SWAY BRACE ASSEMBLY
28 REQUIRED



CARLOAD DATA

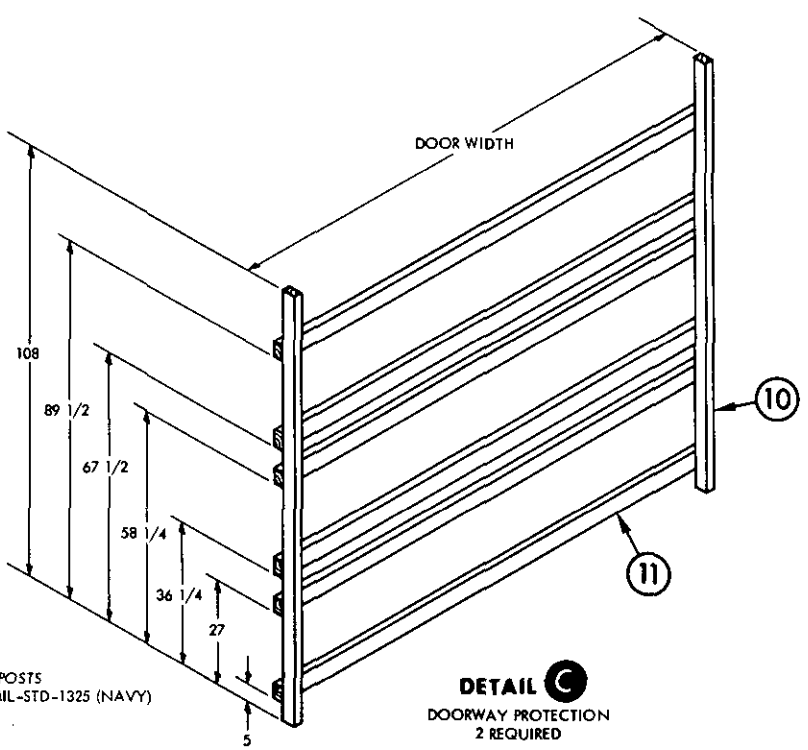
NUMBER OF UNIT LOADS	84
LOAD WEIGHT	50, 820 LBS
DUNNAGE WEIGHT	1, 692 LBS
CARLOAD WEIGHT	52, 512 LBS

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*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL B
CENTER GATE
2 REQUIRED



WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

DETAIL C
DOORWAY PROTECTION
2 REQUIRED

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